



Semester 1 Examinations 2012 / 2013

Exam Code(s)	1SD1, 1MF1, 1SD3
Exam(s)	Higher Diploma (Software Design & Development) M.Sc. (Software Design & Development) Higher Diploma (Software Design & Development) – Industry Stream
Module Code(s)	CT516 CT861 CT538 CT542
Module(s)	Computer Architecture and Operating Systems Architecture, Operating Systems and Networks for .NET Architecture, Operating Systems and Networks for Unix
Paper No.	I
External Examiner(s)	Prof. Michael O’Boyle
Internal Examiner(s)	Professor Gerard Lyons Dr. Michael Madden *Dr. Michael Schukat *Dr. Jim Duggan *Dr. Enda Howley
<u>Instructions:</u>	CT516 and CT861: Answer 2 questions from Section A and one question from section B. CT538 and CT542: Answer 2 questions from Section A and one question from section C. .
<i>Duration</i>	2 hrs
No. of Pages	7 (Including cover)
Discipline(s)	Information Technology
Requirements	None

OLLSCOIL NA hÉIREANN
NATIONAL UNIVERSITY OF IRELAND, GALWAY

SEMESTER I, WINTER 2012-2013

Higher Diploma (Software Design & Development)
M.Sc. (Software Design & Development)
Computer Architecture and Operating Systems (CT861 and CT516)

Higher Diploma (Software Design & Development) – Industry Stream
Architecture, Operating Systems and Networks for .NET (CT538)
Architecture, Operating Systems and Networks for Unix (CT542)

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Time Allowed: 2 hours

CT861 and CT516: Answer 2 questions from Section A and one question from section B.

CT538 and CT542: Answer 2 questions from Section A and one question from section C.

Section A – All Students (Answer 2 Questions)

Question 1

- a) Distinguish between *distributed*, *embedded*, *desktop* and *mobile computers*. Where can these types of computers be found in the area of *everyday activities*? Provide one example per computer type.

10 MARKS

- b) Distinguish between *CISC* (Complex Instruction Set Computer) and *RISC* (Reduced Instruction Set Computer)?

5 MARKS

- c) Show how a single assignment (i.e. $p = p + 2$; with p being an integer variable) can be translated into a set of
- RISC instructions
 - CISC instructions
- for a hypothetical CPU.

10 MARKS

Question 2

- a) Assume you launch a standard application like MS Word on your PC. Using an example outline how and under which conditions the launched process goes through the various states of the *7-state process model*.

10 MARKS

- b) What is the idea of the *stored program concept* as introduced by John von Neumann? How does this concept fit into the *von Neumann architecture*?

9 MARKS

- c) What are in your view the advantages and disadvantages of the *Harvard architecture* over the *von Neumann architecture*?

6 MARKS

Question 3

- a) All modern CPUs have *interrupt* support. Why is this important? Use a detailed example (outlining the features and benefits of interrupt processing) to illustrate your answer.

10 MARKS

- b) Assume you design a CPU with 16-bit instructions where each instruction has 2 operand references. Draw up a *suitable* instruction format and calculate how many instructions and operand references you can distinguish.

5 MARKS

- c) Distinguish between the concept of *pre-allocation* and *dynamic allocation* in secondary storage management. Describe one pre-allocation algorithm and one dynamic allocation algorithm and outline their advantages and disadvantages.

10 MARKS

Question 4

- a) A compiler translates a high level program into a sequence of CPU-specific instructions. Provide code fragments (written in a high-level programming language of your choice) that are translated into the following CPU instruction types:

- Data transfer
- Arithmetic
- Logical
- Conversion
- Transfer of control

Provide one example per instruction type to illustrate your answer.

10 MARKS

- b) Distinguish between the terms *paging* and *demand paging*. Provide an example that shows the advantage of the latter over the former.

7 MARKS

- c) What is meant by term *locality of reference*? How does it relate to the concept of *caching* and to *cache design*?

8 MARKS

Section B - CT516 and CT861 (Answer 1 question)

Question 5

- a) Using examples distinguish between the different types of *active* and *passive* networks attacks.
10 MARKS
- b) Give guidelines for the production of strong but memorable passwords.
5 MARKS
- c) A hacker tries to gain access to your internet-enabled computer. What *types of malware* could (s)he use to achieve this? Explain your answer.
10 MARKS

Question 6

- a) Outline the benefits and problems of the following disk scheduling policies:
- FIFO
- SSTF
- SCAN
- C-LOOK.
10 MARKS
- b) What are the *disk scheduling requirements* for multimedia data? Explain your answer.
6 MARKS
- c) Using a detailed example show how a *disk cache* can boost hard disk performance.
9 MARKS

Section C - CT538 and CT542 (Answer 1 question)

Question 7

- a) Describe in detail the four layers of the TCP/IP Model.

13 MARKS

- b) A sample client-to-server HTTP GET message is the following:

```
GET /kurose_ross/interactive/quotation8.htm HTTP/1.1
Host: gaia.cs.umass.edu
Accept: text/plain, text/html, text/xml, image/gif, image/jpeg, audio/mp4,
audio/mpeg, video/wmv, video/mpeg, application/*, */*
Accept-Language: en-us, en-gb;q=0.5, en;q=0.2, fr, fr-ch, de, ar, cs
If-Modified-Since: Fri, 26 Oct 2012 01:35:47 EDT
User Agent: Mozilla/5.0 (Macintosh; Intel Mac OS X 10_7_3)
AppleWebKit/534.55.3 (KHTML, like Gecko) Version/5.1.5 Safari/534.55.3
```

- What is the name of the file that is being retrieved in this GET message?
- What version of HTTP is the client running?
- What languages is the browser indicating that it is willing to accept?
- What is the client's preferred version of English? What is the browser's least preferred version of English?
- Does the client already have a (possibly out-of-date) copy of the requested file? Explain. If so, approximately how long ago did the client receive the file, assuming the GET request has just been issued?
- What is the type of client browser and the client's operating system?

12 MARKS (2 MARKS EACH)

Question 8

a) Discuss each of the following, with the use of examples to illustrate your answers.

- Peer-to-peer file distribution
- SMTP
- DNS
- UDP

13 MARKS

b) Explain using code examples how you can implement sockets in Python. You need to show both client and server perspectives.

13 MARKS